BE-SAFE – BicyclE SAFEty for Mobility and Active Riding in Smart Cities

STARTING GRANTS FORM

1. Please select one of the following themes.

Note: You can only choose ONE of the themes below to apply for a Starting Grant.

Choose one of the following answers

Intelligent Information Processing for the Digital Transition: Promises and Risks

X Green and Secure ICT for the digital society in Anthropocene

2. Title of the project

BE-SAFE – BicyclE SAFEty for Mobility and Active Riding in Smart Cities

3. Participant details (individuals, institutions)

Non-academic partners

- *Town Hall of the Municipality of Pamplona*
- *IRIS Digital Innovation Hub (Navarra)*
- *Town Hall of the Municipality of Covilhã*
- *ANSR - Portuguese Authority of Road Safety*
- *ESRI Portugal - Environmental Systems Research Institute*
- *Province of Brescia*
- *Municipality of Brescia*
- Brașov Metropolitan Agency
- ACS Kronstadt Cycling (Romania

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Role in the institution: Assistant Professor, Vice-President of the Department of Civil Engineering and Architecture

UNITA partner 2 - Team Leader

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UNITA partner 3 - Team Leader

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UNITA partner 4 - Team Leader

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4. Proposal

Abstract

Cyclist safety is a growing concern in urban mobility, particularly in the context of smart cities that promote sustainable and active transportation. Despite increasing bike usage, infrastructure and traffic systems often remain inadequately adapted, leading to preventable accidents and reduced adoption of cycling. The BE-SAFE project (BicyclE SAFEty for Mobility and Active Riding in Smart Cities) addresses this critical issue by developing a data-driven, technology-enabled framework to assess and improve cyclist safety.

Through a multidisciplinary collaboration among four UNITA universities, the project will build a data-driven decision-support framework to assess cycling risks, model intervention scenarios, and guide urban infrastructure planning. The approach combines accident data analysis with GIS and machine learning to detect risk patterns and factors, psychological analysis of cyclists' behaviour and perceived needs, IoT-enabled cyclist behaviour tracking via sensorized bike-sharing systems and V2X (Vehicle-to-Everything) communication, and simulation of safety interventions in multimodal traffic environments. By articulating the collective knowledge, this project will specifically identify the primary challenges in enhancing bicycle sensing and cycling safety, fostering the development of more extensive future initiatives.

Expected outcomes include the design of a unified decision-support framework and initial testing to evaluate the potential to produce policy recommendations for municipalities and user guidelines, ultimately fostering safe and sustainable cycling practices.

BE-SAFE's modular and interdisciplinary design ensures that the methods and tools developed can be adapted and scaled to other cities and regions, supporting safer cycling infrastructures globally and contributing to smart and greener urban mobility strategies.

Project's objectives, activities and results

Objective

To develop integrated approaches based on data analysis, sensing technologies, and simulation to improve cyclist safety and optimize active mobility in smart cities.

Specific objectives:
1. Identify urban cycling risk factors and accident patterns.
2. Analyze cognitive, emotional, and perceptual influences on cyclist safety and decision-making.
3. Monitor cyclist behavior and interactions via sensorized bikes and V2X communication within a Smart City platform.
4. Simulate and assess urban mobility interventions for safety.

5. Generate actionable insights for infrastructure planning and public policy.

The BE-SAFE project is structured around four interrelated tasks, each contributing to the development of a comprehensive framework for improving cyclist safety and promoting active mobility in smart cities.

Task 1: Data collection and analysis of bicycle-related accidents, followed by predictive risk modelling using ML and GIS **(UBI)**

This task focuses on gathering and analysing historical accident data involving bicycles in urban settings. Public databases, police reports, and mobility datasets will be collected to identify high-risk areas and recurring accident patterns. Using machine learning techniques, predictive models will be developed to estimate the probability of injury severity based on spatial, temporal, and contextual features such as traffic volume, infrastructure characteristics, weather conditions, pedestrian, vehicle and driver information. The results will produce dynamic risk maps to inform future intervention strategies.

Task 2: Cyclist Psychology & Behavior Analysis **(UniTBV)**

This task will focus on designing and administering a structured survey to explore psychological factors like perceived safety, attention, stress levels, behavior under pressure, and trust in infrastructure. The goal is to correlate these psychological variables with reported cycling behaviors to identify risk profiles, motivational factors, and areas where behavioural interventions or infrastructure improvements could be most impactful.

Task 3: Deployment of sensors in bike-sharing fleets to collect behavioural data and test V2X communication **(UPNA)**

This task involves the installation of sensors on selected bike-sharing fleets to collect real-time behavioural data, such as speed, acceleration, braking patterns, and proximity to vehicles or obstacles. In addition, Vehicle-to-Everything (V2X) communication technologies will be tested to evaluate interactions between cyclists, the surrounding traffic and the road infrastructure, enhancing situational awareness, by means of hybrid geometric stochastic simulations, in order to retrieve frequency/power and time domain characteristics of current CV2-X and future Joint Communication and Sensing systems. The collected data will be used to model cyclist behaviour under varying traffic and environmental conditions and identify potentially hazardous situations in real-time, providing input towards the urban dynamic studies developed in Task 4. An orchestrated Smart City platform will be designed, in order to visualize in a comprehensive way the information as well as to provide data processing capabilities.

Task 4: Design and simulate different urban mobility interventions using the outcomes obtained in tasks 1, 2 and 3. **(UniBS - support from UBI, UniTBV and UPNA)**

The goal of Task 3 is the identification of possible strategies to achieve holistic simulations that include multidisciplinary perspectives of Smart Mobility systems. Often simulators look at problems from the angle of a single discipline, in our case, for instance, mobility.

Starting from the data and models on safety and behavior developed in Tasks 1, 2, and 3, this activity will sketch the architecture of a simulation tool for urban mobility scenarios that integrate safety interventions, behaviors, sensing information, and communications. The base platform for such a simulation tool will be the already mentioned SUMO-Veins-PLEXE framework. The multidisciplinary contributions may include: i) the redesign of the road infrastructure, ii) the presence of communication networks (V2X) and sensing; iii) behavioral models for Vulnerable Road Users (VRU); and iv) residual accidents severity.

Task 5: Unified framework and initial validation (**UBI - support from UPNA, UniBS and UniTBV).**

The outputs from Tasks 1 to 4 will be integrated into a unified framework for risk assessment and decision support. This task will validate the approach with small-scale pilot controlled laboratory testing.The aim is to understand the potential of the framework in providing recommendations and user guidelines for city planners, transport agencies and mobility operators. An open-access joint scientific publication will disseminate the project's findings, ensuring the transferability of methods and tools to other urban contexts.

Expected Results
Unified framework for risk assessment and decision support.
Joint scientific publication and follow-up proposal.

Methodology (Max 3500 characters - including spaces)

The BE-SAFE project will adopt an integrated and interdisciplinary approach combining data analysis, psychological profiling, sensing technologies, and simulation to improve cyclist safety in smart urban environments. The process begins with the collection and analysis of historical accident data involving cyclists in urban areas. This involves gathering information from public datasets, police reports, and mobility records to identify risk factors, crash hotspots, and recurring patterns. Advanced machine learning techniques and GIS tools will be used to develop predictive models that estimate the accident probability and severity based on spatial, temporal, and contextual variables. The outcomes of this stage will be dynamic risk maps, which will form the basis for subsequent activities.

Parallel to the data analysis, a psychological and behavioural study will be conducted to understand the influence of cognitive, emotional, and perceptual factors on cycling behaviour and perceived safety. A structured survey assessing aspects such as attention, stress levels, trust in infrastructure, and decision-making under pressure will be designed and applied to cyclists. By correlating these psychological variables with self-reported behaviours and attitudes, the project will identify behavioural risk profiles and motivational patterns to inform the design of behavioural and infrastructural interventions.

To complement the subjective insights from the survey and the patterns identified in the accident data, real-world behavioural data will be collected by deploying sensors in selected bike-sharing fleets. These sensors will monitor variables such as speed, acceleration, braking, and proximity to vehicles or obstacles. In parallel, V2X (Vehicle-to-Everything) communication

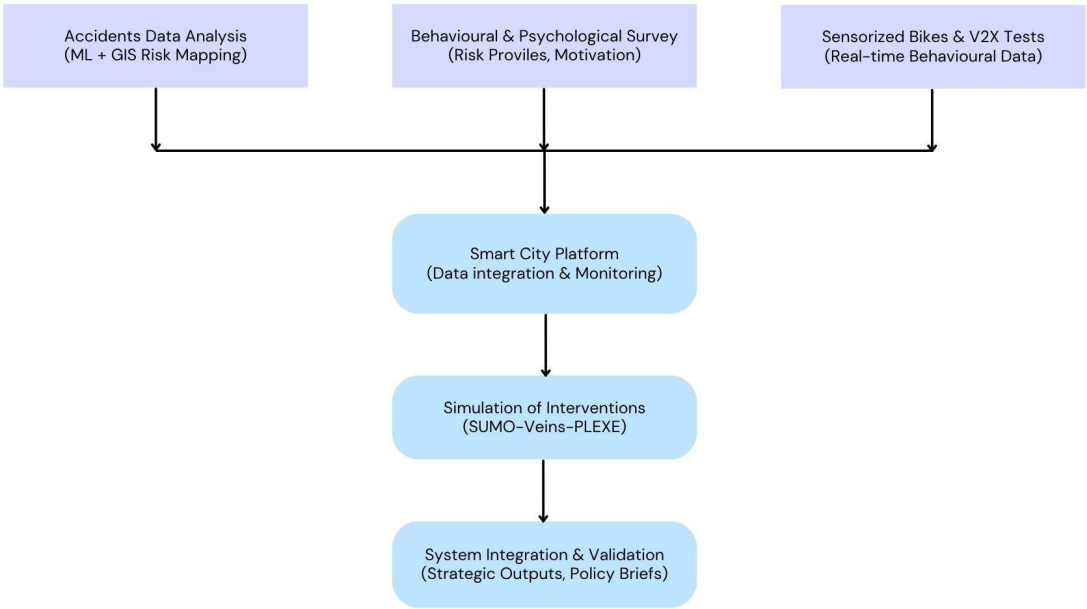
technologies will be tested to capture interactions between cyclists, vehicles, and infrastructure. The system will also include hybrid geometric-stochastic simulations to extract communication and sensing parameters, particularly in the context of current and emerging CV2-X and Joint Communication and Sensing technologies. The data collected will enable detailed behavioural modelling in different traffic and environmental conditions and will be integrated into a visual Smart City platform designed to present the information clearly and intuitively.

Based on these insights, the project will then simulate and evaluate various urban mobility interventions. A simulation architecture will be developed using the data and models generated from previous phases to assess the impact of safety strategies on urban cycling dynamics. This platform will integrate aspects such as infrastructure redesign, communication systems, sensor input, and behavioural models of Vulnerable Road Users. A multidisciplinary approach that goes beyond conventional mobility simulations, including safety, perception, and real-time interaction will be enabled by the simulation framework, which will be based on the SUMO-Veins-PLEXE platform.

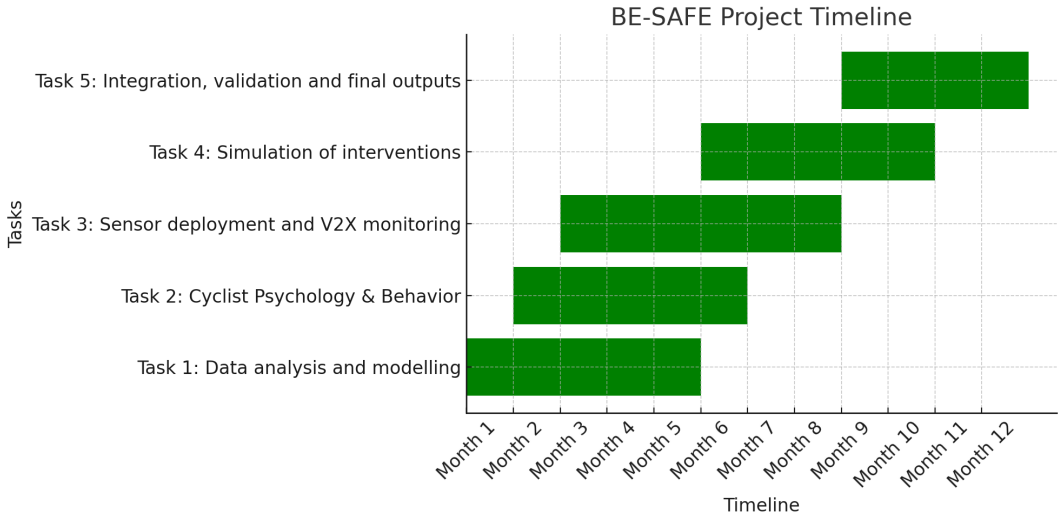
Finally, all components will be integrated into a unified framework for risk assessment and decision support. Its initial validation will occur through small-scale pilot controlled laboratory testing, allowing for an understanding of the framework's potential in defining recommendations and user guidelines. Conclusions drawn may then be tested on a larger scale in future projects to support the creation of safer cycling infrastructures.

The project will conclude with an open-access scientific publication to ensure the methodology is widely disseminated and can be replicated.

BE-SAFE Methodology flowchart



Timeline of the BE-SAFE project



Interdisciplinary approach (Max 3500 characters - including spaces)

The BE-SAFE project takes a deeply interdisciplinary approach, combining complementary scientific domains to tackle the complex challenges of cyclist safety and active mobility in urban environments. BE-SAFE brings together transportation engineering, computer science,

telecommunications, psychology & behaviour, and urban planning. This comprehensive approach links data analysis with policy design to ensure urban cycling safety.

Transportation engineering provides the foundation for understanding urban mobility patterns, accident dynamics, and infrastructure design. Using data modeling and simulation, this field evaluates safety interventions and redesigns public spaces for active transport. Engineering principles are applied to assess the impact of elements such as bike lanes, traffic signals, lighting, and road geometry on cyclist behaviour and safety.

Computer science and machine learning play a key role in converting raw mobility and accident data into actionable insights. Through supervised and unsupervised learning algorithms on large datasets, BE-SAFE identifies hidden patterns, risks and severity, and builds predictive models for cyclist incidents severity. Computer science also supports decision-support systems and integrates diverse data streams.

Real-time monitoring relies on telecommunications and the Internet of Things (IoT). The project collects data on cyclist movement, vehicle interactions, and environmental conditions by deploying V2X communication and sensor-equipped bikes in bike-sharing fleets. This allows near-miss events to be detected and behavioral responses to traffic to be assessed, allowing for proactive risk mitigation.

Cyclist Psychology & Behavior Analysis provides a valuable human-centered perspective, examining how cognitive, emotional, and perceptual factors affect cyclists' decision-making processes, their risk perception, and their overall safety. This involves conducting structured surveys on topics such as perceived safety, attention, stress, behavior under pressure, and trust in infrastructure. The aim is to correlate these psychological variables with reported cycling behaviors to identify risk profiles, motivational factors, and areas in which effective behavioral or infrastructural interventions could be implemented.

Urban and transport planning provides a systemic perspective, ensuring that interventions are aligned with the broader goals of sustainable development, equity, and public health. Planning expertise translates data insights into policies, spatial interventions, and recommendations that promote inclusive and safe mobility. It also ensures that solutions are feasible and scalable, and that they align with municipal strategies and stakeholder needs.

By bridging these domains, BE-SAFE establishes a multi-layered framework in which engineering rigour, technological innovation, behavioral insight, and policy relevance converge. This interdisciplinary integration is essential for the project's scientific robustness, societal impact, and ability to be scaled up in different urban contexts.

The project team includes civil engineers specialized in Transportation, Urban Planning, and GIS, who have experience in applying supervised ML algorithms to transport and safety data. Telecom and computer science engineers contribute with their knowledge of developing orchestrated IoT smart city platforms and designing joint communication and sensing paradigms.

Academic partnership – roles of the partners and partnership development (Max 3500 characters - including spaces)

Roles of the partners

The BE-SAFE scheme unites four academic institutions with complementary knowledge, allowing a cooperative structure that combines data science, behavioural analysis, sensing technologies, mobility systems, and policy translation.

U1 - The work on data-based modelling will be led by UBI, drawing on their experience in accident data analysis, machine learning, and simulation. U1 is responsible for compiling and processing historical cyclist-related incident datasets, developing predictive models to identify high-risk zones, risk-factors, and injury severity to define urban safety strategy and intervention scenarios through simulation.

U2 - UPNA contributes with advanced knowledge in behavioural monitoring and sensor deployment and its integration within an orchestrated Smart City platform. The UPNA has experience in this type of integration within the city of Pamplona in the framework of the H2020 Stardust Project. Its role involves the instrumentation of bicycles in bike-sharing systems with IoT sensors, capturing cyclist behaviour (e.g., speed, braking, interactions, …) under real traffic conditions and considering the main impairments at the physical layer for communication and sensing purposes. This data will be fundamental for understanding human factors and informing real-time risk assessments. Moreover, applied and computational EM resources will also be employed in order to support joint communication and sensing capabilities in collaboration with UniBS.

U3 - UniBS research team, part of the Advanced Networking System group (https://ans.unibs.it) has a long experience in Vehicular Networks, Cooperative Driving applications, and simulation of complex Smart Mobility scenarios, including behavioral models. Within Task 4, UniBS will ensure that the different simulation models and techniques available in the project consortium are homogenized and proper interfaces are designed to let different simulators exchange data and results to enhance the quality of models and predictions.

U4 – UniTBV led the Psychology & Behavior Analysis of cyclists, leveraging expertise in Educational Psychology and Human–Machine Interaction. They will design and conduct studies (observations, surveys, focus groups) to map motivational drivers, communication cues and interaction patterns, producing behavioural models and user guidelines that fine-tune context-aware V2X features and boost system acceptance and safety. Moreover, UniTBV has extensive experience in reproducing bicycle - vehicle accident scenarios under controlled laboratory conditions - to validate safety mechanisms and optimize system robustness.

Supported by UPNA, UniBS, and UniTBV, UBI will lead the development of an unified framework (for tasks 1–4) and initial small-scale pilot validation. This involves developing testing protocols and overseeing the initial pilot laboratory tests. UBI will also analyse framework performance alongside partners to guide future improvements and policy recommendations

aligned with user needs and regulations.Conclusions drawn may then be further developed by the participant universities in larger-scale future projects.

This collaboration is based on complementary research agendas and a mutual commitment to sustainable urban mobility. The project will entail regular coordination, joint publications, and the collaborative supervision of early-career researchers. This integrated academic network will facilitate cross-disciplinary research, enhance knowledge transfer, and ensure cohesive project activities through shared infrastructure.

Sustainability of the project (Max 3500 characters - including spaces)

The BE-SAFE project has been designed with inherent sustainability in mind, ensuring that its outcomes will improve urban cyclist safety and sustainable mobility well beyond the project's lifetime. This commitment is reflected in the project's modular design, open-source approach, collaborative framework, and strategic dissemination plans.

The project's modular and interdisciplinary design is key to its adaptability and longevity. By developing a data-driven framework, BE-SAFE creates tools such as risk models, sensor protocols, and simulations that can be adapted and scaled to other cities around the world. This flexibility maximises widespread adoption and sustained impact. Crucially, all tools and platforms developed will be open source and scalable, promoting broader implementation and continuous improvement by enabling municipalities and researchers to use and adapt them freely.

Emphasising preventive interventions and data-driven decision-making is a core aspect of BE-SAFE's sustainability. The project provides municipalities with the tools they need to proactively identify risks and simulate the impact of interventions, resulting in more efficient resource allocation and a reduced need for costly corrective actions. This creates a cycle of continuous improvement and minimises waste.

The collaborative academic partnership between UBI, UPNA, UniBS, and UniTBV significantly contributes to sustainability by fostering cross-disciplinary research and the collaborative supervision of early-career researchers, thereby building long-term capacity. Joint publications, workshops, and datasets ensure that knowledge is disseminated widely. The team will actively pursue further funding (e.g. from UNITA and Horizon Europe) to ensure the continuation and expansion of the project's work.

Finally, BE-SAFE's strategic focus on active collaboration with municipalities and providing actionable insights ensures its relevance in the real world. We will collaborate closely with Pamplona's Smart City initiative, a city already involved in cross-regional projects such as Stardust, in which the UPNA team developed a Smart City platform for green mobility. This direct engagement provides a valuable testing ground for our solutions and facilitates their integration into ongoing urban development. We will also collaborate with the IRIS Digital Innovation Hub to experimentally validate our technologies and explore opportunities for joint proposal development within the consortium. Our strong connections with key urban innovation

stakeholders ensure that our research translates into practical, implementable solutions, which will significantly boost the project's long-term impact on making cities safer and smarter.

Engagement with non-academic ecosystem (Max 3500 characters - including spaces)

Engagement with municipalities, bike-sharing providers, cycling associations and road safety authorities is central.

Since 2018, UBI has been working with the Municipality of Covilhã on the development of sustainable and intelligent mobility solutions, with a particular focus on cycling strategies for hilly cities. Since 2023, UBI has also had an institutional agreement with the Portuguese Road Safety Authority (ANSR). This agreement provides access to detailed Portuguese road accident data, enabling the development of analyses to support data-driven safety interventions (see document 2 in Section 5). UBI also has an agreement with ESRI Portugal (Environmental Systems Research Institute), a world leader in geographic information systems (GIS), location intelligence, and mapping. ESRI technology provides valuable insights by creating sophisticated maps and performing advanced geospatial analysis.

UPNA has been working with the municipality of Pamplona in the Smart City platform initiative since 2017, within the framework of the H2020 Lighthouse Stardust Project (https://stardustproject.eu/). Moreover, team members of the current proposal are actively engaged with the IRIS Digital Innovation Hub (https://www.irisnavarra.com/en), within the framework of the European Innovation Hub network, coordinating the IoT and Connectivity vertical.

UniBS is part of the Italian National Center for Sustainable Mobility (MOST - https://www.centronazionalemost.it/) and Lombary Mobility Cluster (CLM - https://clusterlombardomobilita.it/). Through these umbrella organizations UniBS cooperates with municipalities, local and regional governments, and has access to entities for road management and industries in the sector.

UniTBV has engaged with the Brașov Metropolitan Agency and ACS Kronstadt Cycling to pilot and validate key components of the system in real-world settings. Through the Brașov Metropolitan Agency, we integrate regional transport planning expertise and data, while collaboration with ACS Kronstadt Cycling ensures that end-user perspectives drive our design. This dual partnership underpins context-sensitive recommendations and maximizes both the impact and transferability of our results.

Workshops, recommendations, and policy briefs will facilitate the engagement with non-academic ecosystems.